on the solenoid valve through the relay module (RM1) and water starts flowing.

The LDR module is used to sense the day and night conditions in the control room. If light intensity in the room is below the threshold level, especially during night, it turns on the light through relay module RM1.

Sensor DHT11 is connected at D6 pin of NodeMCU to monitor the temperature and humidity levels.

Vibration sensor VS1 is connected to D8 pin of NodeMCU. Any vibration near the plant(s) is registered as unwanted activity, which alerts the user through the glowing of inbuilt LED in NodeMCU board.

NodeMCU has only one analogue input (A0 pin) but we may require as many as six sensors to interface with MCU. This problem can be overcome by using 16x1 analogue/digital high-speed CMOS multiplexer 74HC4067 with NodeMCU-ESP32.

Soil moisture sensors SS1 through SS4 and one LDR sensor module are connected to input pins C0 through C4 of 74HC4067 module. The status of the five analogue inputs from these five sensors is selected using select lines S3, S2, S1, and S0. These select lines control by D3, D2, D1 and D0 pins of NodeMCU. Their connection details are listed in Table 1. The list of sensors connected in the system are shown in Table 2.

After selecting the input from 74HC4067 multiplexer module through the selection lines, a single output from the multiplexer available at pin SIG is connected to analogue input A0 of NodeMCU. The NodeMCU controller processes the input signals and turns on the light or activates the solenoid and connects the signals to the Internet.

The DHT11 sensor's data output is connected to D6 pin of NodeMCU for detection of temperature and humidity.

To control the AC-voltage-operated lights and solenoid valve for water supply to plants, D7 and D5 pins of NodeMCU are connected to input pins IN2 and IN1 of the relay module, respectively.

Software

Arduino IDE is used for programming the NodeMCU board. Connect the NodeMCU to PC/laptop and select proper COM port and board name from Tools menu of the Arduino IDE. The board name used in this project is shown in Fig. 4.

Fig. 2: Author's prototype

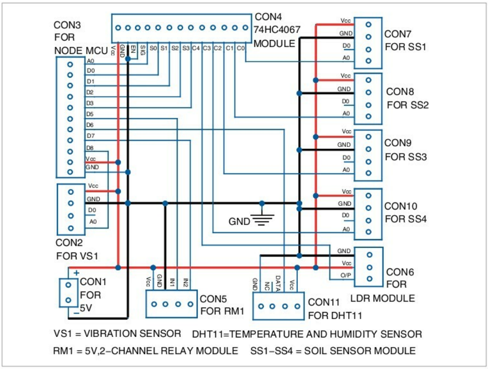

Fig. 3: Circuit diagram of the smart agriculture system

Pin Connections Between 74HC4067 and NodeMCU and Selection Condition of Pins — Table 1

SN pins of 74HC4067 connected to DN pins of NodeMCU				Select Input
D3→S3	D2→S2	D1→S1	D0→S0	SIG/A0
LOW	LOW	LOW	LOW	C0
LOW	LOW	LOW	HIGH	C1
LOW	LOW	HIGH	LOW	C2
LOW	LOW	HIGH	HIGH	C3
LOW	HIGH	LOW	LOW	C4

Where S1 through S3 are selection lines and D0 through D3 are digital pins